

$$ud \frac{1}{16 \pi^2} \left(\frac{1}{324} \frac{1}{m_e^2} (\overline{y_d}^{pi3} (27 \overline{y_u}^{ri1} (-y_d^{ri4} y_u^{pi2} + 6 s_\gamma^2 c_\gamma^2 y_d^{pi4} y_u^{ri2}) - 2 g_1^2 s_\gamma^2 y_d^{pi4} \delta_{i1i2}) + \right.$$

$$\begin{aligned} & 7 g_1^2 c_\gamma^2 \overline{y_u}^{pi1} y_u^{pi2} \delta_{i3i4}) - \frac{8}{243} \sum_p g_1^4 LF_{3,0}[m_d^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{5}{81} \sum_p g_1^4 LF_{4,-1}[m_d^p] \delta_{i1i2} \delta_{i3i4} - \frac{32}{1215} \sum_p g_1^4 LF_{5,-2}[m_d^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{8}{81} \sum_p g_1^4 LF_{3,0}[m_e^p] \delta_{i1i2} \delta_{i3i4} + \frac{5}{27} \sum_p g_1^4 LF_{4,-1}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{32}{405} \sum_p g_1^4 LF_{5,-2}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} \sum_p g_1^4 LF_{3,0}[m_l^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{5}{54} \sum_p g_1^4 LF_{4,-1}[m_l^p] \delta_{i1i2} \delta_{i3i4} - \frac{16}{405} \sum_p g_1^4 LF_{5,-2}[m_l^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{4}{243} \sum_p g_1^4 LF_{3,0}[m_q^p] \delta_{i1i2} \delta_{i3i4} + \frac{5}{162} \sum_p g_1^4 LF_{4,-1}[m_q^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{16}{1215} \sum_p g_1^4 LF_{5,-2}[m_q^p] \delta_{i1i2} \delta_{i3i4} - \frac{32}{243} \sum_p g_1^4 LF_{3,0}[m_u^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{20}{81} \sum_p g_1^4 LF_{4,-1}[m_u^p] \delta_{i1i2} \delta_{i3i4} - \frac{128}{1215} \sum_p g_1^4 LF_{5,-2}[m_u^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{54} (\overline{y_d}^{pi3} (9 \overline{y_u}^{ri1} (y_d^{ri4} y_u^{pi2} (c_\gamma^2 - s_\gamma^2)^2 + 6 s_\gamma^2 c_\gamma^2 y_d^{pi4} y_u^{ri2}) - 4 g_1^2 s_\gamma^2 y_d^{pi4} \delta_{i1i2}) + \\ & 2 g_1^2 c_\gamma^2 \overline{y_u}^{pi1} y_u^{pi2} \delta_{i3i4}) LF_{1,2}[m_\Phi] + \\ & \frac{1}{18} (s_\gamma^2 \overline{y_d}^{pi3} (3 c_\gamma^2 \overline{y_u}^{ri1} (2 y_d^{ri4} y_u^{pi2} - 3 y_d^{pi4} y_u^{ri2}) + 2 g_1^2 y_d^{pi4} \delta_{i1i2}) + \\ & g_1^2 c_\gamma^2 \overline{y_u}^{pi1} y_u^{pi2} \delta_{i3i4}) LF_{2,1}[m_\Phi] - \\ & \frac{1}{162} g_1^2 (18 s_\gamma^2 \overline{y_d}^{pi3} y_d^{pi4} \delta_{i1i2} + (9 c_\gamma^2 \overline{y_u}^{pi1} y_u^{pi2} + 8 g_1^2 \delta_{i1i2}) \delta_{i3i4}) LF_{3,0}[m_\Phi] + \\ & \frac{5}{54} g_1^4 LF_{4,-1}[m_\Phi] \delta_{i1i2} \delta_{i3i4} - \frac{16}{405} g_1^4 LF_{5,-2}[m_\Phi] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{4}{81} g_1^4 LF_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{27} g_1^4 LF_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{32}{405} g_1^4 LF_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{243} g_1^4 LF_{2,1,0}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{243} g_1^4 LF_{2,2,-1}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{2}{243} g_1^4 LF_{3,1,-1}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{243} g_1^4 LF_{4,1,-2}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{243} g_1^4 LF_{2,1,0}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{243} g_1^4 LF_{2,2,-1}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{2}{243} g_1^4 LF_{3,1,-1}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{243} g_1^4 LF_{4,1,-2}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{4}{243} g_1^4 LF_{2,1,0}[m_1, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{243} g_1^4 LF_{2,2,-1}[m_1, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{8}{243} g_1^4 LF_{3,1,-1}[m_1, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{243} g_1^4 LF_{4,1,-2}[m_1, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{4}{243} g_1^4 LF_{2,1,0}[m_1, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{243} g_1^4 LF_{2,2,-1}[m_1, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{8}{243} g_1^4 LF_{3,1,-1}[m_1, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{243} g_1^4 LF_{4,1,-2}[m_1, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{4}{81} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{8}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{4}{81} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{8}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{4}{81} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{8}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{4}{81} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{8}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{2}{243} g_1^4 LF_{2,1,0}[m_d^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{243} g_1^4 LF_{3,1,-1}[m_d^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{8}{81} g_1^2 g_3^2 LF_{2,1,0}[m_d^{i3}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_d^{i3}, m_3] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{2}{243} g_1^4 LF_{2,1,0}[m_d^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{243} g_1^4 LF_{3,1,-1}[m_d^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{8}{81} g_1^2 g_3^2 LF_{2,1,0}[m_d^{i4}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_d^{i4}, m_3] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{27} g_1^2 (-2 \overline{y_d}^{pi3} y_d^{pi4} \delta_{i1i2} + \overline{y_u}^{pi1} y_u^{pi2} \delta_{i3i4}) LF_{2,1,0}[m_q^p, \tilde{\mu}] + \\ & \frac{1}{54} g_1^2 (2 \overline{y_d}^{pi3} y_d^{pi4} \delta_{i1i2} - \overline{y_u}^{pi1} y_u^{pi2} \delta_{i3i4}) LF_{2,2,-1}[m_q^p, \tilde{\mu}] + \\ & \frac{1}{54} g_1^2 (2 \overline{y_d}^{pi3} y_d^{pi4} \delta_{i1i2} - \overline{y_u}^{pi1} y_u^{pi2} \delta_{i3i4}) LF_{3,1,-1}[m_q^p, \tilde{\mu}] + \\ & \frac{8}{243} g_1^4 LF_{2,1,0}[m_u^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{4}{243} g_1^4 LF_{3,1,-1}[m_u^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{8}{81} g_1^2 g_3^2 LF_{2,1,0}[m_u^{i1}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_u^{i1}, m_3] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{8}{243} g_1^4 LF_{2,1,0}[m_u^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{4}{243} g_1^4 LF_{3,1,-1}[m_u^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{8}{81} g_1^2 g_3^2 LF_{2,1,0}[m_u^{i2}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_u^{i2}, m_3] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{54} g_1^2 (2 \overline{y_d}^{pi3} y_d^{pi4} \delta_{i1i2} - \overline{y_u}^{pi1} y_u^{pi2} \delta_{i3i4}) LF_{2,1,0}[\tilde{\mu}, m_q^p] + \\ & \frac{1}{54} g_1^2 (14 \overline{y_d}^{pi3} y_d^{pi4} \delta_{i1i2} + 11 \overline{y_u}^{pi1} y_u^{pi2} \delta_{i3i4}) LF_{3,1,-1}[\tilde{\mu}, m_q^p] - \\ & \frac{2}{27} g_1^2 (\overline{y_d}^{pi3} y_d^{pi4} \delta_{i1i2} + \overline{y_u}^{pi1} y_u^{pi2} \delta_{i3i4}) LF_{4,1,-2}[\tilde{\mu}, m_q^p] + \\ & \frac{2}{81} g_1^4 LF_{2,1,1,-1}[m_1, m_d^{i3}, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{4}{81} g_1^4 m_1^2 LF_{2,1,1,0}[m_1, m_d^{i3}, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{2}{81} g_1^4 LF_{2,1,1,-1}[m_1, m_d^{i4}, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{4}{81} g_1^4 m_1^2 LF_{2,1,1,0}[m_1, m_d^{i4}, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{9} g_3^4 LF_{2,1,1,-1}[m_3, m_d^{i3}, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{2}{9} g_3^4 m_3^2 LF_{2,1,1,0}[m_3, m_d^{i3}, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{9} g_3^4 LF_{2,1,1,-1}[m_3, m_d^{i4}, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{2}{9} g_3^4 m_3^2 LF_{2,1,1,0}[m_3, m_d^{i4}, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{6} \overline{y_d}^{pi3} y_d^{ri4} \overline{y_u}^{ri1} y_u^{pi2} LF_{2,1,1,-1}[\tilde{\mu}, m_q^p, m_q^r] - \\ & \frac{1}{3} \tilde{\mu}^2 \overline{y_d}^{pi3} \overline{y_u}^{ri1} (y_d^{ri4} y_u^{pi2} - 3 y_d^{pi4} y_u^{ri2}) LF_{2,1,1,0}[\tilde{\mu}, m_q^p, m_q^r] \end{aligned}$$